Planet Wars Competition Entry

ISILab Team
Alvaro Chocobar Belmar
University of Talca

May 14, 2026

Agent Overview

- **Monte Carlo Tree Search (MCTS)** with domain-specific heuristics:
  - Dynamic exploration-exploitation balance based on game progress and relative advantage
  - Lightweight state evaluation to guide the search
  - Progressive bias to prioritize strategically promising actions during tree expansion
- Key idea: The agent adapts its behavior depending on the phase of the game and advantage over the opponent, prioritizing exploration at the beginning and exploitation at the end.

System Design

- The agent is structured into three main components:
  - **GameStateWrapper**
    - Adapts the game engine to MCTS requirements
    - Handles simulation via Forward Model
    - Generates and evaluates actions
  - **Monte Carlo Tree Search**
    - Selection using a dynamic UCT with guided by progressive bias
    - Expansion
    - Rollout with random policy
    - Backpropagation of heuristic results
  - **Parameter Tuning**
    - Hyperparameters optimised offline with irace (Iterated F-Race)

Results

Win Rate vs Baselines V1

Opponent	Win Rate (%)	Games Played
Greedy Heuristic Agent	50.0	50
EvoAgent-200-20-0.5-true	24.0	50
Overall Average	**37.0**	**100**

Win Rate vs Baselines V2

Opponent	Win Rate (%)	Games Played
Careful Random Agent	100.0	50
Greedy Heuristic Agent	90.0	50
Overall Average	**95.0**	**100**

Notable Matchups

- Second evaluation carried out with the reference agents of the competition
- Monte Carlo with progressive bias noticeably improves the agent's performance

Analysis and Insights

Failure Modes / Observed Behaviour

- The agent is reactive rather than proactive, allowing the opponent to accumulate more troops
- MCTS with UCT guided by progressive bias significantly improves performance

Design Insights

- The agent's actions are no longer structurally guided
- The exploration constant adapts in each child node depending on the estimated advantage over the opponent
- Progressive bias guides the agent's expansion in the early stages of the game towards promising actions, and at the end of the game it works again only with UCT

Future Work

Limitations

- The opponent simulation continues to be based on a passive policy, putting the agent at risk against an aggressive agent

Planned Improvements

- Improve opponent's simulation making it a semi-random roll out
- Further optimize the agent by refining parameter configuration and reducing sources of noise in action generation